Carbon scenario analysis planning

Analysis notes for paper drafting | Updated 20 October 2026

This note records the scenario component planned for the paper and the decisions needed before calculation. No new projections have been agreed or calculated. The intended deliverable is a transparent Excel workbook that colleagues can use without R.

Purpose and available evidence

Explore how measured tree carbon accumulation and soil-carbon information might inform explicit land-use or management scenarios. The annual stock, PAIC and MAIC summaries and complete-profile soil summaries are available as inputs. Historical scenario sheets remain as context; they are not inputs to the current preparation pipeline.

Methods to agree before building

First specify geography, eligible land categories, the area-data source and reference year. Define whether the hectare refers to the whole system or the tree component, and distinguish eligible land from adopted land. Specify adoption through time, projection horizon, hybrid composition, management and stand-age cohorts.

Choose how PAIC, MAIC and stock observations support each calculation. PAIC is an interval growth rate, MAIC is cumulative stock divided by age, and stock is an accumulated amount. They cannot be interchanged as annual rates. A constant-rate extrapolation would need explicit justification and limits.

Keep tree and soil contributions separate until their area and time conventions match. Any soil contribution requires a justified counterfactual and accumulation timescale. A cross-sectional stock difference must not automatically be divided by stand age and presented as a measured sequestration rate.

Current results and discussion

There are no new scenario results to report. Potential interpretation should be framed as conditional outcomes under stated assumptions, not a forecast of achieved removal. Avoid double counting stock and annual increments, tree-row and alley areas, or annual and cumulative totals.

Workbook requirements

Use clearly identified editable inputs, units beside each input, transparent formulas, source references and separate annual and cumulative outputs. If converting carbon to carbon dioxide, state the molecular-weight conversion explicitly. Include uncertainty or sensitivity analysis for the assumptions that materially affect conclusions.

Agree scenario labels, baseline or counterfactual, eligible-area fractions, adoption schedules, hybrid-specific growth trajectories, soil treatment, and the handling of harvesting, mortality and other carbon pools before producing paper figures. Avoid adding unsupported precision to uncertain inputs.

Files supporting this note

scenarios/README.md

data/processed/annual_hybrid.csv

data/processed/annual_hybrid_treatment.csv

data/processed/soil_stock_summary.csv

data/raw/soil_and_legacy_scenarios.xlsx

Scenario clarification 21 October 2026

The following decisions update the earlier planning sections. They specify the intended calculation without reporting new scenario results.

Methods and agreed scope

Replay the observed approximately 30-year hybrid-specific age trajectory for each future planting cohort. This is the intended forward scenario, rather than extrapolation of a single PAIC or MAIC value. Allocate multi-year PAIC intervals consistently across their elapsed years. PAIC describes interval-average accumulation; age-specific MAIC reconstructs accumulated stock when multiplied by the corresponding age. Check consistency against observed stocks using common records and age definitions.

Present AGB C separately as the principal diameter-based allometric estimate, accompanied by combined stock = AGB C + estimated BGB C + ESM-corrected SOC. Estimate BGB C at each age as R times AGB C at that age, retaining the same 50 percent carbon fraction for both biomass pools. Retain R = 0.26 provisionally; 0.20 and 0.40 are proposed sensitivity assumptions, not confidence limits. Keep absolute stocks separate from additional stocks relative to the reference. Mortality, harvest and maintenance remain discussion considerations; the calculation assumes standing, maintained trees through the scenario horizon.

Proposed soil comparison

Recover soil-profile locations and link them to original cropping-treatment plots, retaining sampling location separately from treatment history. In 2023, Continuous fallow alleys may represent approximately 30 years since conversion from arable, while former Continuous cropping alleys may represent approximately 20 years since cessation. Confirm exact dates before calculating durations. Handle Alternate cropping separately because its cultivation history differs.

For each conversion-history group, estimated additional SOC = mean ESM alley SOC minus comparable mean ESM arable SOC. Divide each contrast by its own years since conversion to obtain an inferred average annual accumulation rate (Mg C per hectare per year). These are two conversion ages sampled on one occasion, not repeated measurements of the same soil. Shared former-field origin strengthens reference plausibility but does not establish identical initial stocks. Tree-row SOC requires separate spatial weighting and management interpretation.

Calculated results

No new projection or conversion-history-specific SOC rate has been calculated. The sample map and exact conversion dates are needed before estimating the two soil contrasts.

Discussion hypotheses

Differences between the two conversion-history groups may inform the time course of SOC accumulation, but also reflect cultivation legacies, spatial variation and different durations of tree influence relative to cessation. A lower 30-year annualised rate alone would not demonstrate saturation. The two contrasts can constrain scenario assumptions; they do not uniquely determine an annual SOC curve. For trees absent at baseline, accumulation of living biomass corresponds to additional tree carbon under the stated baseline and retention assumptions.

Publication to review

HIGHLIGHTED LITERATURE NOTE: Schulz-Nielsen, L., Burmeister, J.-M., Fiege, C., Richter, R. and Pecenka, R. (2026). Management Effects on Biomass Partitioning in Fast-Growing Poplar in Brandenburg. Forests 17(3), 395. Published 23 March 2026. https://doi.org/10.3390/f17030395

Check whether the thesis bibliography already includes this publication. It provides directly sampled aboveground and belowground biomass, including coarse and fine roots, and supports testing uncertainty in root allocation. Important correction to the earlier chat description: the study concerns stands in their sixth and seventh growing years, not 30-year-old trees. Its ratio estimates should therefore support a provisional assumption rather than be presented as validation for older stands. Management and coppicing affect the ratio; do not pool post-harvest ratios indiscriminately with unharvested stands.

Open decisions

Obtain the sample map, treatment assignments, conversion dates and available spatial replication. Verify the new publication against the thesis bibliography. Decide how the two soil contrasts constrain the scenario trajectory, including early years; retain uncertainty and avoid attributing all SOC gain to trees alone. Confirm cohort timing, eligible area, geometry and hybrid selections before calculation.